A Multilevel and Robust Investigation of E-commerce Delivery Lead Time

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Abstract

This report applies Advanced Multivariate Statistics methods—multilevel modeling, model-based clustering, robust estimation, and bootstrap inference—to the analysis of delivery lead times in Brazilian e-commerce. Using the Olist dataset (2016–2018, final modelling sample $n = 93{,}853$), I address three research questions: (RQ1) variance decomposition across order, seller, and state levels; (RQ2) identification of key routing and shipment drivers; and (RQ3) robustness of findings to outliers and modelling choices. Results suggest that seller and customer-state heterogeneity remains non-negligible in the pooled model; in M3, the incremental same-state term is about -29%, and the implied total same-state contrast versus cross-region shipments is about -35%. Extending the baseline model with shipment-regime clusters ($K = 5$) and interaction terms (M4_c2) improves fit (AIC 117189 $\rightarrow$ 115806; marginal R^2 0.222 $\rightarrow$ 0.247) and indicates heterogeneous routing/freight effects across regimes. Robustness checks (trim95, robust flat estimators, regional refits, and bootstrap intervals) preserve effect directions for reported terms and support a stable directional inferential story.

Keywords: multilevel models, robust statistics, bootstrap inference, delivery lead time, hierarchical data

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1 Introduction

1.1 E-commerce delivery lead times

Over the last decade, e-commerce has shifted from a niche channel to a core retail format in many economies, with online marketplaces acting as intermediaries between thousands of dispersed sellers and millions of customers.(Fleischmann et al., 2008; Gunasekaran et al., 2002) As product choice, prices, and user interfaces converge across platforms, delivery performance has become a key competitive lever: slow or unreliable shipping directly affects customer satisfaction, reviews, and repeat purchase behaviour.(Xiao et al., 2017; Boyer et al., 2005) Customers not only expect their orders to arrive, but to arrive within increasingly tight and accurately communicated delivery windows.

In large and geographically diverse markets such as Brazil, achieving reliable delivery performance is particularly challenging. Delivery lead times can vary widely across regions and sellers, reflecting long distances between major cities and remote areas, heterogeneous transport infrastructure, and differences in operational practices across thousands of independent sellers.(Duque, 2023) Understanding how much of the observed variation in delivery lead time is due to geography, seller heterogeneity, and shipment characteristics is therefore important both for platform operations (e.g. setting service-level agreements and matching rules) and for individual sellers seeking to improve their logistics performance. Publicly available transaction data from the Olist Brazilian e-commerce platform provide a rich setting to study such variability at scale, with detailed order-level information on timestamps, locations, and shipment attributes.(Olist and Kaggle Contributors, 2018)

1.2 Research questions and statistical challenges

In this study, I use the Olist data as a case study to apply Advanced Multivariate Statistics Inference (AMS) methods to a real-world hierarchical problem in e-commerce logistics. From a statistical perspective, this type of dataset raises two main challenges.

First, orders are clustered: multiple observations come from the same sellers and destination regions, so the independence assumption[1] underlying ordinary least squares is often violated, which can lead to underestimated standard errors and inflated Type-I error rates if clustering is ignored. (Asampana Asosega et al., 2024; Hox and Schoot, 2013) Multilevel models or linear mixed-effects models are explicitly designed for such grouped data and allow me to decompose variance into within- and between-cluster components, quantify intraclass correlations, and estimate cluster-specific effects with shrinkage toward the overall mean.[2]

Second, delivery lead times are strongly right-skewed with a long upper tail, and a non-negligible fraction of observations behave like outliers. In this setting, classical maximum-likelihood or REML estimators can yield unstable fixed-effect estimates and standard errors, motivating robust sensitivity checks (trimmed variants and robust flat estimators) alongside multilevel modelling. (Finch, 2017; Koller, 2016) In addition, shipment characteristics are multivariate, correlated, and heavy-tailed, which makes it natural to consider model-based clustering of shipment profiles as a way to identify latent shipment regimes and test whether routing and freight effects are homogeneous across these regimes.

The analysis is organised around three research questions, phrased explicitly in statistical terms:

RQ1 (variance decomposition). How much of the variability in delivery lead time is attributable to differences between orders, between sellers, and between customer states?

RQ2 (key effect drivers). Within this multilevel framework, how are key geographic and shipment characteristics (such as same-state shipping, freight cost, item count, and order value) associated with faster or slower deliveries?

RQ3 (robustness and stability). To what extent are the main findings stable when we change modelling assumptions and stress-test the data?

[1] The independence assumption states that, conditional on the covariates in the model, the error terms of different observations are uncorrelated. In clustered data (e.g. multiple orders from the same seller or region), within-cluster errors are typically positively correlated, so treating them as independent leads to underestimated standard errors.

[2] This is partial pooling: groups with less information are pulled more strongly toward the global mean, while well-observed groups are pulled less.

1.3 Project goals and report outline

The primary goal of this project is pedagogical rather than managerial: it uses the Olist case to show how methods from the Advanced Multivariate Statistics (AMS)course can be combined to analyse clustered, skewed data in practice. Methodologically, the project illustrates (i) when and how to specify random-effects structures for hierarchical data, (ii) how to interpret variance components, intraclass correlations, and fixed effects in mixed models, and (iii) how robust estimators and bootstrap procedures can be used to assess the stability of conclusions under outliers and model misspecification.

The remainder of the report is structured as follows. Section 2 reviews how AMS approaches have been used in applied business analytics, and highlights gaps in their joint application to business problems. Section 3 introduces the Olist dataset and key features of the analysis sample. Section 4 outlines the modelling framework and empirical strategy used to analyse delivery lead times. Sections 5.2, 5.5 and 5.6 present the main empirical results for RQ1–RQ3, covering variance decomposition, key effect drivers, and robustness checks. Finally, Sections 6 and 7 interpret the findings, discuss methodological implications and limitations, and close with the main contribution of the study.

This project is expected to contribute a reproducible template for students and analysts facing similarly structured hierarchical business data, showing how AMS methods can be combined in a transparent, end-to-end analysis of e-commerce delivery lead times.

2 Literature Review

2.1 Background of Multivariate Statistics approach

Multivariate statistics grew out of the need to study how several predictors jointly relate to one or more outcomes rather than looking at variables in isolation.(*An Introduction to Multivariate Statistical Analysis* 2026) Researchers in fields such as economics, psychology, and business are often interested in questions like “which factors jointly drive this outcome?” or “how do groups of variables move together in real data?”. Classical linear and multivariate models answered these questions under relatively strong assumptions about the data.

In practice, however, real datasets are rarely that clean. Observations are often clustered (for example, customers within regions or orders within sellers), a small fraction of cases behave as clear outliers, and there may be hidden subgroups that follow different patterns from the bulk of the data. At the same time, outcomes and predictors can be skewed or heavy-tailed, and relationships may vary systematically across groups. Under these conditions, a single pooled linear model with textbook formulas for standard errors tends to underestimate uncertainty, be overly influenced by extreme points, and smooth over meaningful between-group differences.

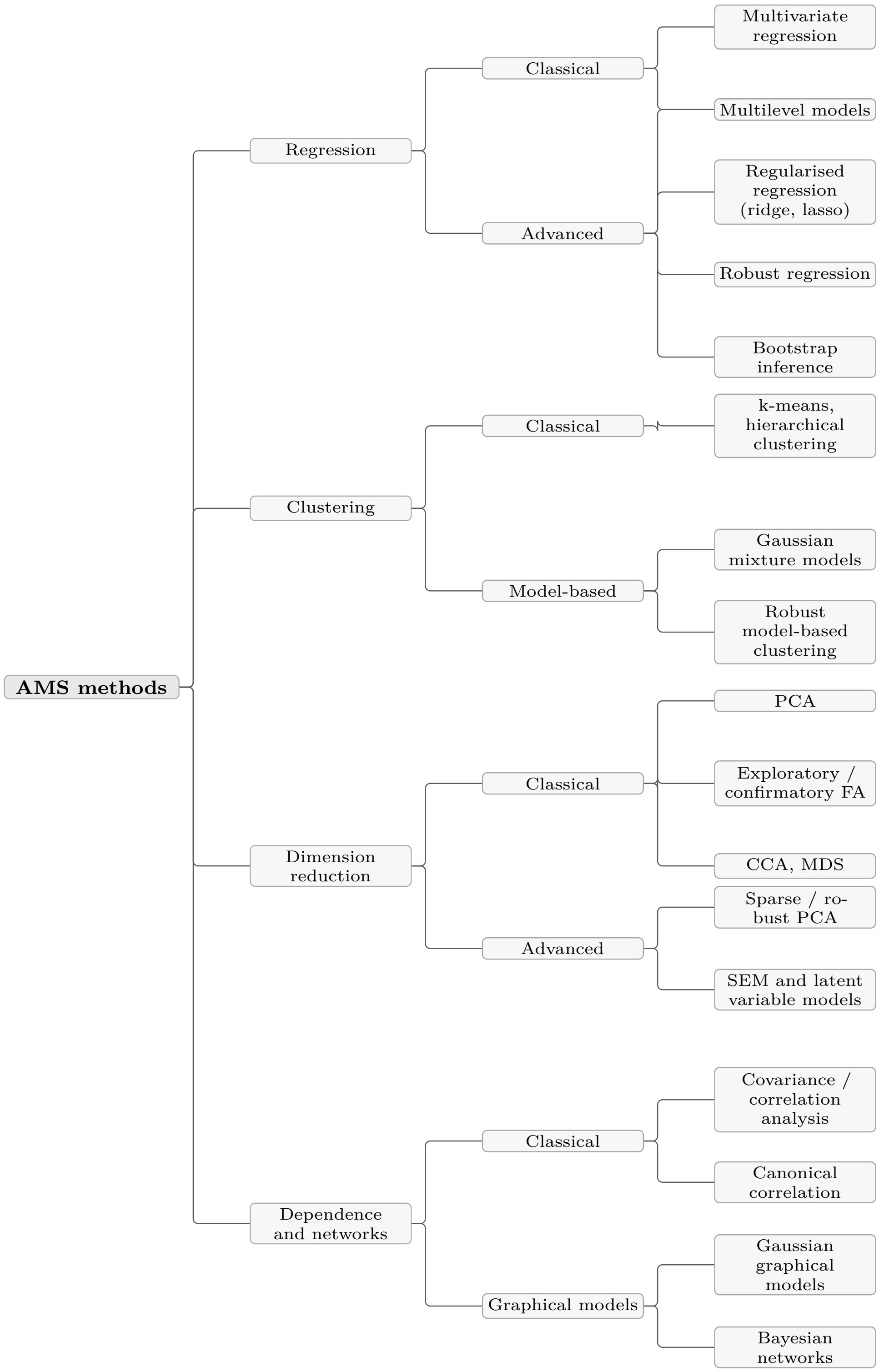
Advanced Multivariate Statistics (AMS) was developed to deal with this kind of complexity. Building on these complications, Figure 1 situates the methods used in this project within the broader landscape of Advanced Multivariate Statistics, organizing them by modelling goal (regression, clustering, dimension reduction, dependence) and by whether they extend classical parametric tools with multilevel structure, robustness, regularisation, or resampling-based inference. (Snijders and Bosker, 2011)

Overall, this structural map of Advanced Multivariate Statistics (AMS) methods highlights the broad families of tools that empirical researchers routinely use to analyse complex, high-dimensional data. In the next subsection, I briefly review how multivariate analysis has been applied in academic empirical research across fields such as psychology, marketing, and operations, before turning to the specific choices made for the Olist delivery data.

2.2 AMS in academic empirical research

In management and organizational research, multilevel modelling has become central for studying cross-level phenomena. Aguinis et al. (2013) documented a substantial increase in multilevel studies while noting common issues such as under-reporting ICCs and unclear distinction between within- and between-group effects. LeBreton and Senter (2008) discussed best practices for aggregation from individual to team or unit levels and emphasized careful multilevel specification for cross-level hypotheses. These papers reinforce the focus on explicit variance decomposition and interpretation of seller- and state-level random effects, even though the application here is descriptive rather than theory-testing.

Figure 1: A structural map of Advanced Multivariate Statistics (AMS) methods



Robust multivariate methods have also been applied across empirical domains to protect against outliers. For example, Hubert et al. (2005) compared classical and robust principal component analysis in chemometric data and showed that robust PCA can recover structure under contamination while classical PCA is distorted. J. Olive (2017) provided practitioner-oriented examples where robust regression and robust covariance estimators changed substantive conclusions in the presence of leverage points. These studies support the decision to complement classical mixed-model inference with trimming and robust estimator checks so conclusions about routing and shipment effects are not artifacts of a small number of extreme orders.

For mixed-effects models specifically, Field and Welsh (2007) compared parametric, residual, and case-resampling bootstrap methods in simulation studies and found that parametric bootstrap generally achieved better coverage for variance components and fixed effects under unbalanced clusters. Carpenter et al. (2003) likewise illustrated bootstrap use in multilevel models to obtain more reliable confidence intervals in educational data. These results directly inform the choice to use parametric bootstrap on the fitted mixed-effects model to obtain percentile intervals for key coefficients and ICCs, and to compare them with classical Wald intervals.

Model-based clustering has been increasingly used in applied work to uncover latent regimes in marketing, finance, and operations data, but often as a stand-alone unsupervised analysis.(Fraley and Raftery, 2002; Maitra, 2010) Fewer studies explicitly integrate mixture-based clusters into subsequent regression or multilevel models as heterogeneity design tools, which is the role it plays in the present report.

2.3 AMS in applied business analytics

Compared with health, education, and organizational research, explicit use of AMS methods in routine business analytics remains less common. Aguinis et al. (2013) noted that many multilevel applications are concentrated in psychology and management and encouraged broader adoption where data are naturally nested. Asampana Asosega et al. (2024) highlighted that even in health sciences, key multilevel quantities such as variance components and ICCs are still not always fully reported. Davidson and MacKinnon (2006) similarly observed that bootstrap methods remain underused in many applied econometric studies, where analysts still rely mainly on asymptotic standard errors. Applied robust-methods reviews such as J. Olive (2017) make a related point: robust procedures are often discussed in specialist contexts but less consistently adopted in everyday applied workflows.

In the e-commerce context, logistics datasets exacerbate the issues that AMS methods are designed to handle. Delivery lead times are typically right-skewed with long upper tails; orders are naturally clustered within sellers, customer regions, and time periods; shipment characteristics are multivariate, correlated, and sometimes heavy-tailed; and cluster sizes are highly unbalanced. Yet much of the empirical logistics and operations literature still relies on simple regressions, descriptive comparisons, or panel models with limited attention to variance components, robustness to outliers, or latent regimes in shipment profiles.

In short, this literature indicates a practical gap: relatively few didactic, end-to-end case studies show how to combine multilevel modelling, robust estimation, bootstrap inference, and model-based clustering on realistic business logistics datasets. This study addresses that gap in a modest, applied way using Olist e-commerce data as a case study.

2.4 AMS integration for this study

Raudenbush and Bryk (2002) popularized hierarchical linear models in educational data, showing that students nested within schools exhibit substantial between-school variance and that ignoring this structure can underestimate standard errors and bias inferences about school-level effects. More recently, Asampana Asosega et al. (2024) reviewed multilevel applications published between 2010 and 2020 and reported frequent use of variance components and intraclass correlation reporting, while also noting gaps in methodological reporting. This supports the decision to use multilevel models to partition variance in delivery lead time across sellers and customer states and to report ICCs as part of the core output.

In robust statistics, Filzmoser et al. (2008) surveyed robust methods for multivariate data and showed that high-breakdown estimators for covariance, regression, and PCA can improve estimation when even a small fraction of observations are outlying. Applied treatments and domain examples (e.g. J. Olive (2017) and Frosch et al. (2005)) similarly emphasize the practical value of downweighting atypical observations so they do not dominate parameter estimates. These findings motivate the use of trimming and robust estimator sensitivity checks in this report to assess whether extreme delivery delays unduly influence fixed effects and variance components.

For bootstrap methods, (Davison and Hinkley, 1997) showed across many regression and time-series examples that bootstrap standard errors and confidence intervals can provide more accurate coverage than asymptotic approximations in small-sample or nonstandard settings. Davidson and MacKinnon, 2006 reached similar conclusions in econometrics, highlighting more reliable inference for coefficients and test statistics when classical assumptions are doubtful. In the AMS course context, Masci (2026) presents bootstrap as a general tool for approximating sampling distributions in regression and multilevel settings when analytic formulas are fragile. These works justify the use of parametric bootstrap to validate mixed-model confidence intervals for key coefficients and ICCs.

Model-based clustering complements these tools by treating heterogeneous multivariate predictor spaces as arising from mixtures of latent regimes. Fraley and Raftery (2002) and Maitra (2010) show that Gaussian mixture models, estimated via the EM algorithm and selected by criteria such as BIC, can recover meaningful clusters in correlated, heavy-tailed data where distance-based methods such as k-means are less reliable. This study uses model-based clustering in that spirit, as a way to detect shipment regimes in the multivariate shipment-feature space before fitting cluster-conditional mixed-effects models.

Taken together, these strands yield a coherent AMS workflow: multilevel models provide variance decomposition and clustered inference on delivery lead times. Table 1 summarizes the AMS strands most relevant for this study and the data issues they address.

Table 1: Advanced Multivariate Statistics Structural Methods

Method family	Primary data issue	Role in this project
Multilevel models	Clustered / hierarchical data	Decompose variance across orders, sellers, and customer states; obtain cluster-adjusted fixed effects.
Robust estimators	Outliers, heavy tails	Check sensitivity of key coefficients to extreme delivery delays and atypical shipment profiles.
Bootstrap inference	Fragile asymptotics, unbalanced clusters	Validate confidence intervals for fixed effects and variance components in mixed-effects models.
Model-based clustering	Latent regimes in multivariate predictors	Identify shipment regimes and test cluster-conditioned routing and freight effects.

Overall, the existing literature on applied Advanced Multivariate Statistics in business problems suggests that reliable modelling of e-commerce lead times requires tools that can handle hierarchical structure, outliers, fragile asymptotics, and latent heterogeneity. In the next section, I describe the Olist dataset and the study design that operationalise these insights for the present context.

3 Data and Insights

3.1 Data sources and quality

The analysis uses the publicly available Olist E-Commerce Public Dataset, which contains transaction records from a Brazilian marketplace spanning September 2016 to October 2018.(Olist and Kaggle Contributors, 2018) The core data in Table 2 consist of six relational tables.

Table 2: Dimensions of raw Olist tables used in the analysis.

Table	Rows	Columns
orders	99,441	8
order items	112,650	7
products	32,951	9
customers	99,441	5
sellers	3,095	4
geolocation	1,000,163	5
category matrix	73	6

Data quality in Table 3 is mostly good: missingness is limited (notably 3% missing customer-delivery timestamps), and key IDs/joins are consistent.

Table 3: Variables with missing values in the Olist datasets

Column	Missing (n, %)	Table
order_delivered_customer_date	2,965 (2.98%)	orders
product_category_name	610 (1.85%)	products
product_name_lenght	610 (1.85%)	products
product_description_lenght	610 (1.85%)	products
product_photos_qty	610 (1.85%)	products
order_delivered_carrier_date	1,783 (1.79%)	orders
order_approved_at	160 (0.161%)	orders
product_weight_g	2 (0.0061%)	products
product_length_cm	2 (0.0061%)	products
product_height_cm	2 (0.0061%)	products
product_width_cm	2 (0.0061%)	products

In addition, a product-category matrix (73 categories) is used to map Portuguese category names to English labels and to classify products as hedonic vs. utilitarian and search vs. experience goods for theme controls. The study period captures typical e-commerce activity in Brazil before major external shocks, with orders concentrated in the Southeast region (especially São Paulo) but covering all 27 Brazilian states. With this initial exploration in place, the next section examines data-quality issues and key empirical patterns in more detail before constructing the final analysis dataset for the modelling stage.

3.2 Exploratory Data Analysis Insights

3.2.1 Hierarchical structure and Data Imbalance

The exploratory analysis indicates a clear hierarchical organisation of the data, together with marked cluster-size imbalances:

- Most sellers handle relatively few orders (median $\approx$ 10–20 orders per seller), while a small number of high-volume sellers account for thousands of orders.
- About 98.7% of orders involve exactly one seller, so the analysis is restricted to single-seller orders to ensure clean seller-level grouping (see Table 4).
- Customer states are highly imbalanced: São Paulo (SP) accounts for roughly 40% of orders, while some northern states have fewer than 100 orders. Figure 2

This imbalance is typical in real-world clustered data and motivates mixed-effects models, which naturally account for unequal cluster sizes through shrinkage. To obtain a clear level-2 grouping for mixed-effects models, the analysis therefore restricts the main analysis to single-seller orders, so that each observation can be uniquely associated with a single seller random intercept.

Table 4: Number of sellers per order and share of multi-seller orders

Distribution of sellers per order	
Number of sellers	**Number of orders**
1	97,388
2	1,219
3	54
4	3
5	2

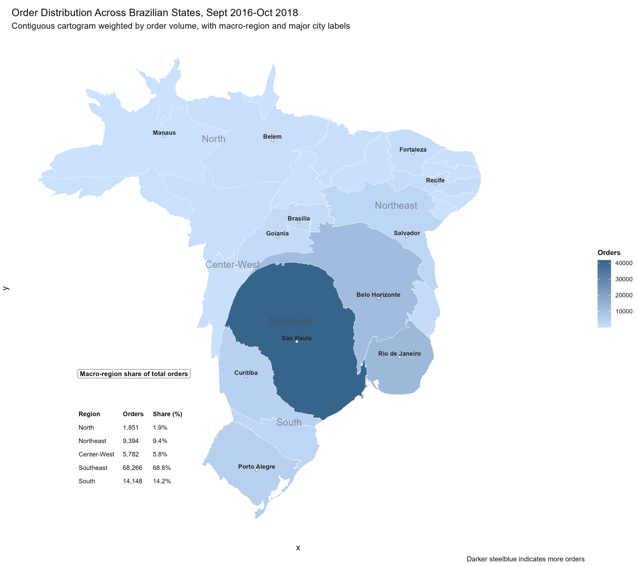


Figure 2: Order distribution map across Brazilian states.

- **Level 1 (orders)**: individual transactions ($n \approx 95{,}000$ after cleaning);
- **Level 2 (sellers)**: unique seller IDs ($n \approx 3{,}000$).

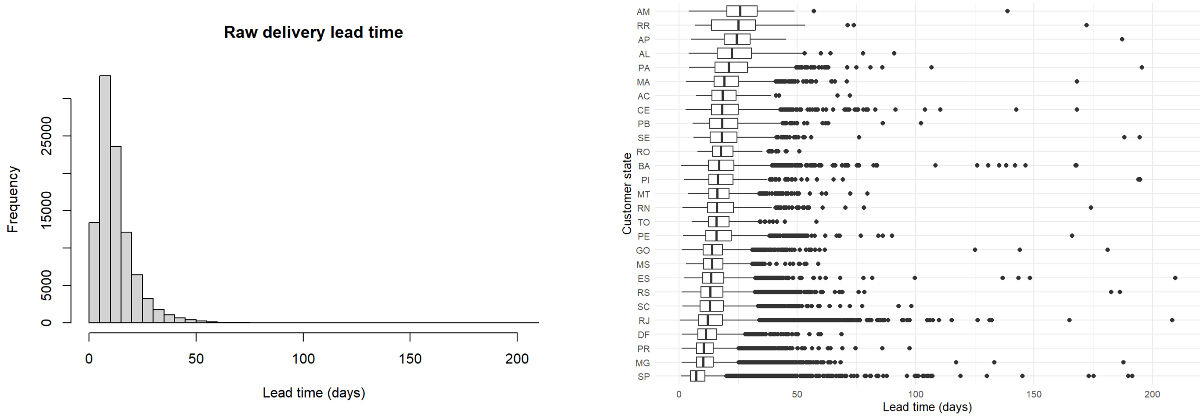
3.2.2 Skewness and heavy tails

Outcome variable To provide a foundation for the modelling choices in Section 4, the analysis examines how delivery times vary across states and sellers. Figure 3b shows average delivery lead time in days by customer state. States in the Southeast, such as São Paulo (SP), exhibit lower median and mean lead times (around 8–9 days), whereas several northern states (e.g., Amazonas, AM, and Pará, PA) have means above 23–26 days and much higher upper quantiles.

Figure 4a shows that geography drives large variation: Southeast is fastest, while North/Northeast are much slower; state-level heterogeneity is substantial.

3.2.3 Correlation between route-level and delivery

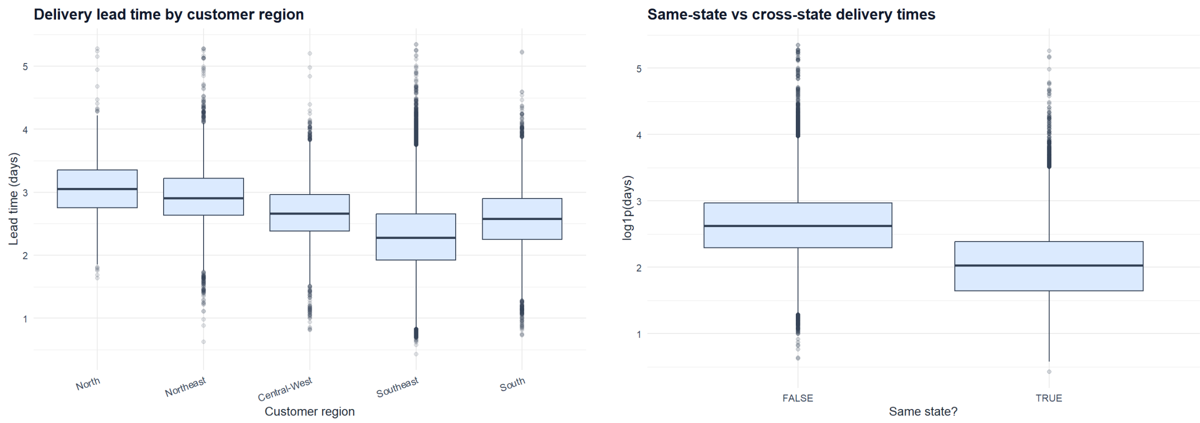
Before multilevel modelling, route-level summaries already show a marked same-state gap. Table 5 compares raw delivery lead times for same-state versus cross-state shipments in the pre-strict single-seller sample ($n = 95{,}194$).



(a) Delivery lead time in days.

(b) Average lead time by customer state.

Figure 3: Outcome distribution and state-level geographic differences.



(a) Lead time by macro-region ($\log(1 + \text{days})$).

(b) Same-state vs cross-state distributions.

Figure 4: Routing and regional contrasts in delivery lead time.

Statistic	Value
Cross-state orders	60,961
Cross-state median lead time (days)	12.841
Cross-state mean lead time (days)	15.208
Cross-state 90th percentile (days)	26.362
Same-state orders	34,233
Same-state median lead time (days)	6.590
Same-state mean lead time (days)	7.967
Same-state 90th percentile (days)	14.280

Table 5: Baseline delivery-time contrast by routing group in the single-seller delivered sample ($n = 95{,}194$).

Figure 4b visualises the same-state advantage reported in Table 5: the full distribution is shifted down for same-state shipments, not only the median.

The Figure 5 chart also shows that correlation between `is_same_state` and `log_lead_time` is -0.493, the strongest univariate predictor in the dataset. Same-state orders have median lead time 6.590 days versus 12.841 days for cross-state orders, with similar separation in means and upper quantiles. This is an unadjusted descriptive contrast, so it may be larger than the conditional M3 coefficient once seller/state heterogeneity and shipment covariates are controlled. It motivates explicit routing terms in the mixed-effects models and provides context for RQ2.

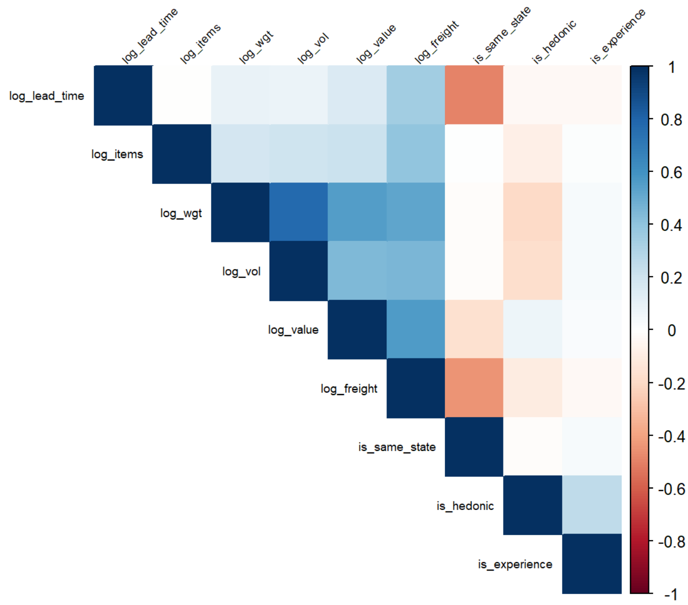
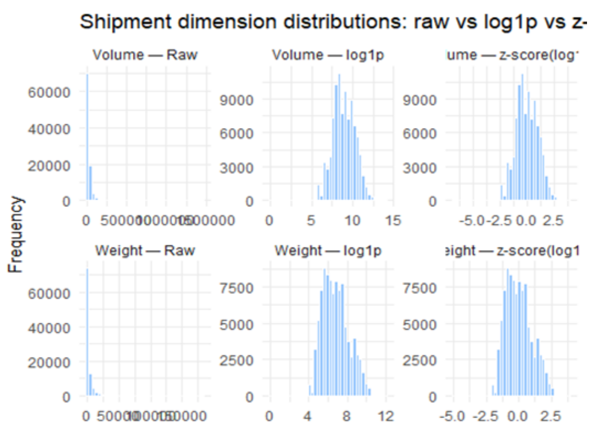


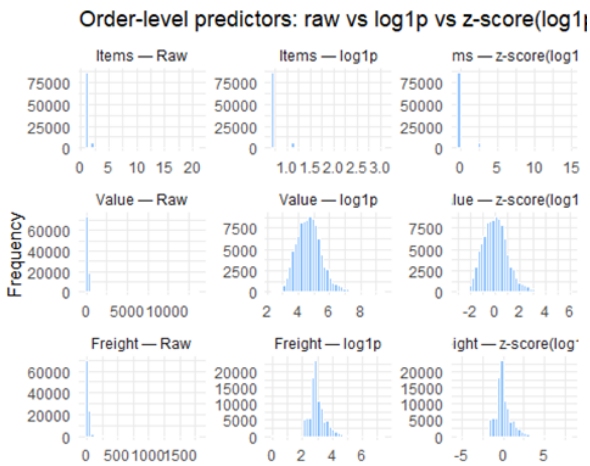
Figure 5: Correlation structure between `log_lead_time` and key predictors.

3.2.4 Multicollinearity

To understand the distributional properties of shipment dimensions, the analysis first compares raw, log-transformed, and standardised versions of volume and weight. In raw form, the Figure 6a and Figure 6b show that both shipment volume and weight are extremely right-skewed, with a long tail of very large consignments and most orders compressed near zero. Applying a log1p transformation yields approximately bell-shaped distributions for both variables, and subsequent standardisation of log1p values produces centred, unit-scale z-scores while preserving this near-symmetric shape. This motivates using z-scored log1p volume and weight as covariates in the multilevel models, rather than their raw counterparts.



(a) Volume and weight dimensions.



(b) Items, value, and freight dimensions.

Figure 6: Shipment-dimension distributions after transformation.

Table 6: Variance inflation factors for shipment covariates

Variable	VIF
log_wgt	3.080682
log_vol	2.563891
log_value	1.704067
log_freight	2.344288
log_items	1.251906
log_customer_pop	1.268755
log_seller_pop	1.080982

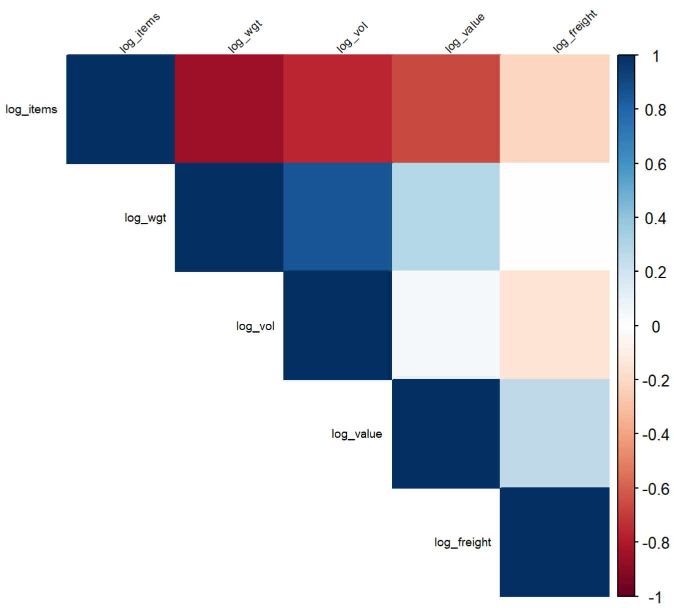


Figure 7: Correlation matrix between shipment variables.

The analysis next examines multicollinearity among the log-transformed shipment covariates. The correlation heatmap in Figure 7 shows strong positive associations between log items, log weight, and log volume, plus weaker but still positive correlations for log value and log freight. The VIFs in Table 6 are modest (about 1.1 to 3.1), so even the most correlated variables can be included together in the multilevel models without problematic inflation of standard errors.

3.2.5 Insights Summary and Implication

The EDA provides a reasonably clear modelling direction. Data quality is sufficient for inference (limited missingness, consistent joins, and a dominant single-seller order structure), while outcome and covariate diagnostics indicate that a standard flat linear model would be fragile.

- **Data quality and hierarchy readiness.** Missingness is limited (about 3% for customer delivery timestamps), joins are coherent, and 98.7% of orders are single-seller; this supports a stable order–seller grouping structure.
- **Outcome distribution.** Lead time is strongly right-skewed with a long upper tail, so log-transformation is required to stabilise variance and reduce leverage from extreme delays.
- **Geographic and routing heterogeneity.** Lead times differ materially across states and regions, and same-state routing is the strongest pre-model signal; geography and routing indicators should remain core predictors.

- **Predictor usability.** Shipment covariates are correlated but not at problematic levels (moderate pairwise correlations and low VIFs), so they can be retained jointly without immediate dimensionality reduction.

These findings directly motivate the strategy used in the next Section 4: crossed multilevel models for clustered structure, transformed outcomes and skewed predictors, and robustness checks (tail capping, robust estimators, and bootstrap inference) to validate directional stability.

3.3 Pre-model multivariate diagnostics (MCD)

Before fitting mixed models, I assess multivariate shipment outliers using robust Mahalanobis distances from the Minimum Covariance Determinant (MCD) estimator on five standardized log shipment predictors (weight, volume, value, freight, and item count).(Filzmoser et al., 2008)

For order i with feature vector $\mathbf{x}_i$, the robust squared distance is:

$$d_i^2 = (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_{\text{MCD}})^\top \hat{\boldsymbol{\Sigma}}_{\text{MCD}}^{-1} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_{\text{MCD}}).$$

Here $\hat{\boldsymbol{\mu}}_{\text{MCD}}$ and $\hat{\boldsymbol{\Sigma}}_{\text{MCD}}$ are the robust MCD location and covariance estimates of the shipment-feature vector. With $p = 5$ features, I flag multivariate outliers using the rule

$$d_i^2 > \chi^2_{p,0.975}.$$

Here $\chi^2_{p,0.975}$ is the 97.5th percentile of a chi-square distribution with p degrees of freedom.

Using this cutoff, about 9.6% of orders are flagged as multivariate outliers in shipment-feature space. This is not used as an automatic deletion rule; instead, it is treated as pre-model evidence that tail/outlier sensitivity is material and should be stress-tested in modelling.

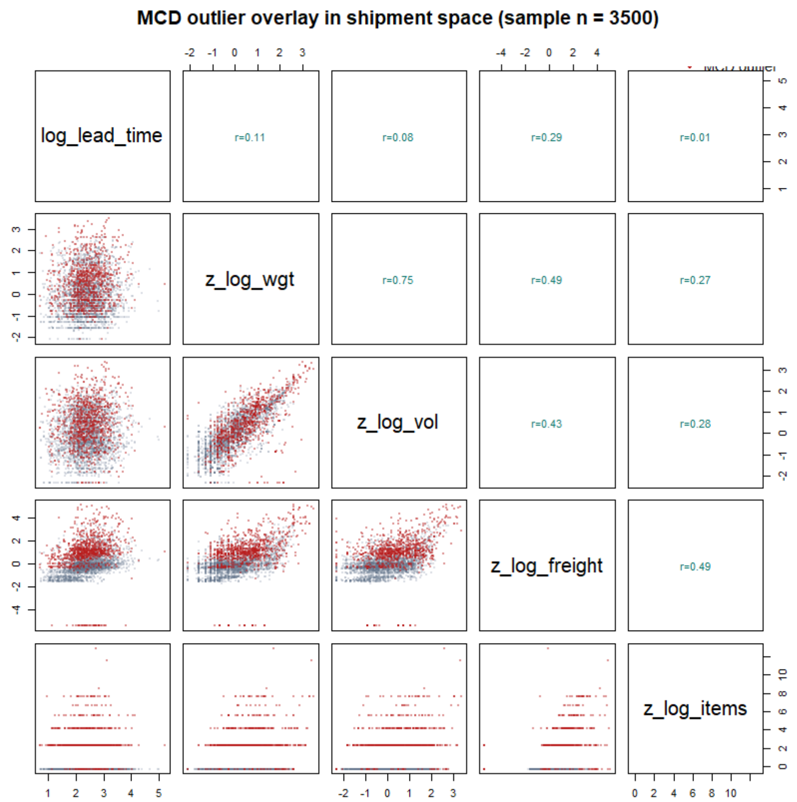


Figure 8: MCD robust multivariate outlier diagnostics on shipment predictors (illustrative pairwise projection).

The flagged points concentrate in extreme shipment combinations rather than around the central cloud, which supports the robustness design in Sections 4.5 and 5.6.

3.4 Final Data Construction

Variable definitions and transformation logic are documented in Section 4 under Sections 4.2.1 and 4.2.2. This subsection focuses on strict sample construction for modelling.

3.4.1 Strict sample filtering

The strict modelling sample is obtained by sequential filtering:

- Start from all orders ($n = 99{,}441$).
- Keep delivered orders with valid purchase and customer-delivery timestamps and nonnegative lead time (`orders_base`, $n = 96{,}469$).
- Keep single-seller orders only (`orders_clean_core`, $n = 95{,}194$).
- Create `orders_model_strict` by removing rows with missing volume, weight, and removing orders without category.

After this strict filter, the final modelling-ready table contains $n = 93{,}853$ orders, 2,892 sellers, and 27 customer states. This is the clean dataset used for the pre-model clustering stage and the main mixed-effects model sequence. In the next section, the modelling strategy, model specifications, and overall workflow are described in detail.

4 Methodology

This section describes the modelling strategy used to answer the research questions and directional expectations developed from Sections 2 and 3. The empirical design combines a crossed multilevel linear mixed-effects model for log delivery lead time with three AMS extensions: robust estimators for outlier sensitivity, parametric bootstrap for interval validation under unbalanced clusters, and model-based clustering to capture latent shipment regimes. Together, these tools are tailored to the hierarchical, skewed, and heterogeneous structure of the Olist data.

4.1 Modelling goals and overview

The primary goal is to explain and decompose variation in delivery lead time across orders, sellers, and customer geographies, while quantifying the roles of routing, shipment characteristics, product context, and seasonality. The modelling strategy is guided by five directional expectations:

H1. Seller and customer-state random-effect variances are non-zero.

H2. Same-state routing has the strongest negative association with log lead time among core fixed effects.

H3. Shipment covariates (freight, items, value) are informative but smaller in magnitude than routing effects.

H4. Cluster-conditioned models reveal heterogeneity in routing and freight sensitivities across shipment regimes.

H5. Core directional conclusions remain stable under tail capping, robust estimators, and bootstrap checks.

4.1.1 Inferential hypotheses and decision rules

To keep hypothesis statements statistically explicit, I define formal null hypotheses where a direct model test is available, and use $\alpha = 0.05$ as the default decision threshold.

- **H1 (clustering):** $H_{0,1a}: \sigma_u^2 = 0$ and $H_{0,1b}: \sigma_v^2 = 0$. In this report, H1 is judged primarily from non-zero variance components and material ICC shares (practical evidence of clustering), since boundary-variance LRTs are not the main inferential device used in the workflow.
- **H2 (routing block):** $H_{0,2}: \beta_s = \beta_r = 0$, tested by nested likelihood-ratio comparison $M0$ vs $M1$.
- **H3 (shipment block):** block null $H_{0,3}^{\text{block}}: \boldsymbol{\gamma} = \mathbf{0}$ conceptually corresponds to $M1$ vs $M2$, while term-level nulls $H_{0,3j}: \beta_j = 0$ are evaluated with Wald tests in $M2/M3$.
- **H4 (regime heterogeneity):** $H_{0,4a}: \boldsymbol{\theta} = \mathbf{0}$ ($M3$ vs $M4\_c1$) and $H_{0,4b}: \boldsymbol{\phi} = \boldsymbol{\psi} = \boldsymbol{\omega} = \mathbf{0}$ ($M4\_c1$ vs $M4\_c2$), both via likelihood-ratio tests on the common cluster-ready sample.
- **H5 (robustness):** no single scalar null is imposed; conclusions are accepted as robust when effect directions and practical magnitudes remain stable across trimming/specification variants, robust estimators, regional refits, and bootstrap-vs-Wald interval checks.

Throughout Results, wording follows these rules explicitly: reject H_0, or fail to reject H_0, only when a direct test is defined.

Operationally, this leads to a crossed mixed-effects framework with random intercepts for sellers and customer states, fixed effects for routing and shipment-related features, and a sequence of robustness and heterogeneity modules (robust estimators, bootstrap inference, and model-based clustering) to stress-test core conclusions.(Gelman and Hill, 2007; Filzmoser et al., 2008; Fraley and Raftery, 2002)

4.2 Variable construction and transformations

All models are built on a common constructed feature set so coefficient changes across M0–M4 reflect model structure, not changing variable definitions.

4.2.1 Outcome transformation

The outcome is order-level delivery lead time in days,

$$\text{lead\_time\_days}_i = t_{\text{deliv},i} - t_{\text{purch},i},$$

where $t_{\text{deliv},i}$ is the customer-delivery timestamp and $t_{\text{purch},i}$ is the purchase timestamp for order i. It is computed for delivered orders in the strict modelling sample. Negative or implausible lead times are removed during data construction (Section 3). The remaining distribution is strongly right-skewed (long upper tail), so the primary modelling scale is

$$\text{log\_lead\_time}_i = \log\bigl(1 + \text{lead\_time\_days}_i\bigr).$$

As shown in Figure 9, this transformation stabilizes variance and improves residual behavior relative to raw-scale fitting. Tail-capped and trimmed variants are reserved for robustness checks (Section 4.5).

4.2.2 Predictors and covariate transformation

Routing and geography Routing contrasts are built from customer and seller locations. Brazilian state codes (UF) are mapped to the five official macro-regions (North, Northeast, Central-West, Southeast, South), and route indicators are then defined as:

- `is_same_state` = 1 if seller and customer are in the same state, 0 otherwise;
- `is_same_region` = 1 if seller and customer are in the same macro-region, 0 otherwise.

Because same-state shipments are always in the same region, `is_same_state` is nested within `is_same_region`. With both indicators in one model:

$$\eta_i = \ldots + \beta_{\text{sr}} R_i + \beta_{\text{ss}} S_i,$$

where η_i is the linear predictor on the log-lead-time scale, R_i = `is_same_region`$_i$, and S_i = `is_same_state`$_i$, while the ellipsis denotes the intercept and other fixed-effect terms not written explicitly.

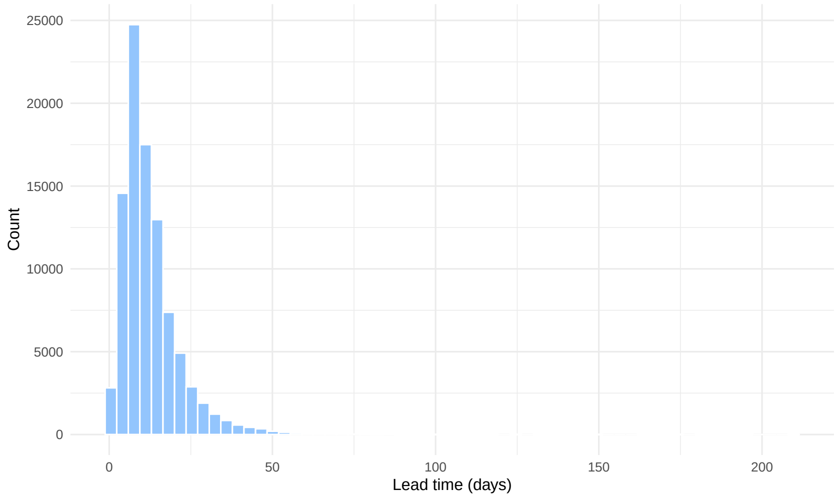


Figure 9: Distributional diagnostics for delivery lead time (raw, transformed, and winsorized scales).

The contrast *same-region (different-state) vs cross-region* is β_{sr}, the *incremental same-state effect* is β_{ss}, and the *total same-state vs cross-region contrast* is $\beta_{\mathrm{sr}} + \beta_{\mathrm{ss}}$. These variables encode the strongest pre-model geographic contrast documented in Section 3 and remain core routing predictors in M1–M4.

Shipment and demand covariates The order-level feature table merges orders, order-items, products, customers, and sellers, then aggregates item attributes to one row per order. Core shipment predictors are `n_items`, `total_value`, `total_freight`, `total_weight_g`, and `total_volume_cm3`. To reduce skewness and make coefficients comparable, positive quantities are transformed by $\log(1 + x)$ and standardized to z-scores: `z_log_items`, `z_log_value`, `z_log_freight`, `z_log_wgt`, and `z_log_vol`. Missing dimension fields are handled by controlled median imputation before aggregation, and strict filtering removes unresolved invalid rows. Correlation/VIF checks indicate moderate, manageable collinearity, so these predictors are retained jointly.

Product-mix controls Each order is assigned a main product category by aggregating item categories at order level and weighting each item by `price + freight`. In the raw mapping there are 74 categories; 22 categories have fewer than 100 orders and together represent about 0.85% of observations. These rare and unknown levels are collapsed into `other`, yielding `main_cat_pt_collapsed` and avoiding unstable tiny-cell estimates.

To retain behavioural interpretation, detailed categories are projected onto two binary dimensions:

- **`is_hedonic`:** hedonic vs utilitarian consumption context.
- **`is_experience`:** experience vs search evaluation context.

Crossing them yields four product-theme quadrants used as conceptual controls for product mix, not causal treatment variables.(*Information and Consumer Behavior | Journal of Political Economy: Vol 78, No 2* 2026; Hirschman and Holbrook, 1982)

- **Search + Utilitarian** (e.g., tools, office goods, basic electronics).
- **Search + Hedonic** (e.g., leisure electronics, hobby categories).
- **Experience + Utilitarian** (e.g., function-oriented apparel and personal care).
- **Experience + Hedonic** (e.g., decor, gifts, fashion accessories).

Additional controls Temporal controls are encoded as fixed factors:

- **Season:** DJF = December–January–February, MAM = March–April–May, JJA = June–July–August, SON = September–October–November.

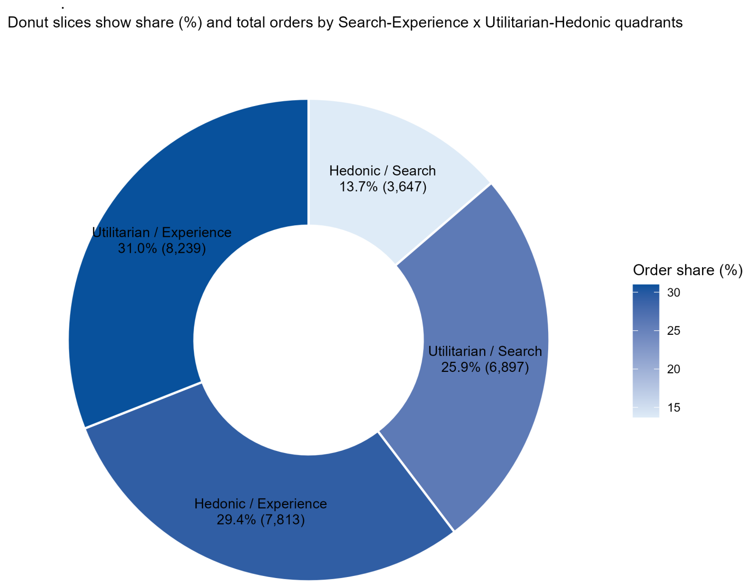


Figure 10: Conceptual Matrix of Product Categories

- **Purchase year:** indicator levels for 2016–2018.

Weekday and popularity features are retained in the engineering table for diagnostics and profiling, but the core pooled mixed-model sequence (M0–M4) uses season, purchase year, and product-theme controls as the time/composition adjustment block.

All subsequent models use this common set of constructed variables unless otherwise noted.

4.3 Baseline multilevel model specification

4.3.1 Crossed random-effects design

The dependence structure is crossed: (1 | seller_id) + (1 | customer_state).[3] This avoids misspecifying the hierarchy as nested and preserves the two major clustering channels seen in EDA.

Operationally, crossed random intercepts capture:

- seller baseline speed differences (fulfillment process and carrier mix),
- destination-state baseline differences (distance and last-mile constraints).

The baseline random-effects formula in `lme4` notation is:

$$\begin{aligned} \texttt{y} \sim \texttt{fixed\_effects} + (1 \mid \texttt{seller\_id}) \\ + (1 \mid \texttt{customer\_state}). \end{aligned}$$

where $\texttt{y} = \texttt{log\_lead\_time}$ and `fixed_effects` denotes the routing, shipment, season/year, and product-theme terms introduced in Section 4.2.

4.3.2 Core mixed model and variance decomposition

Let y_{ijk} denote log lead time for order i, seller j, and customer state k. The baseline mixed model is:

$$y_{ijk} = \beta_0 + \mathbf{x}_{ijk}^\top \boldsymbol{\beta} + u_j + v_k + \varepsilon_{ijk}, \tag{1}$$

with $u_j \sim \mathcal{N}(0, \sigma_u^2)$, $v_k \sim \mathcal{N}(0, \sigma_v^2)$, and $\varepsilon_{ijk} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, assuming random effects are independent of each other and of $\mathbf{x}_{ijk}$.(Gelman and Hill, 2007) Here β_0 is the fixed intercept, $\boldsymbol{\beta}$ is the fixed-effect coefficient

[3] Crossed means seller and customer-state are modeled simultaneously as non-nested group factors. Sellers serve multiple states, and each state receives orders from many sellers.

vector, and $\mathbf{x}_{ijk}$ includes routing indicators (`is_same_state`, `is_same_region`), standardized shipment covariates (`z_log_wgt`, `z_log_vol`, `z_log_value`, `z_log_freight`, `z_log_items`), and time/product controls (`season`, `purchase_year`, `theme_name`).

For variance decomposition (H1), I report:

$$\begin{aligned} \mathrm{ICC}_{\text{seller}} &= \frac{\sigma_u^2}{\sigma_u^2 + \sigma_v^2 + \sigma_\varepsilon^2}, \\ \mathrm{ICC}_{\text{state}} &= \frac{\sigma_v^2}{\sigma_u^2 + \sigma_v^2 + \sigma_\varepsilon^2}, \end{aligned} \tag{2}$$

$$\mathrm{ICC}_{\text{total}} = \frac{\sigma_u^2 + \sigma_v^2}{\sigma_u^2 + \sigma_v^2 + \sigma_\varepsilon^2}, \tag{3}$$

where $\mathrm{ICC}_{\text{total}}$ is the overall clustering share. Models are estimated with restricted maximum likelihood (REML) for variance-component stability; nested fixed-effect comparisons are run with maximum likelihood (ML) refits.

In `lme4` notation, the general fixed-effects backbone used across M2–M4 can be summarized as:

$$\begin{aligned} \texttt{y} \sim\ & \texttt{routing + shipment + season} \\ & + \texttt{purchase\_year + themes} \\ & + (1 \mid \texttt{seller\_id}) \\ & + (1 \mid \texttt{customer\_state}). \end{aligned}$$

4.4 Model progression (M0–M4)

Model complexity is added stepwise to track fixed-effect changes, random-effect variance shares, and fit gains under a constant crossed random-effects backbone. To keep the setup readable, the progression is summarized in a structured design table instead of long algebraic expansions. In every model, the outcome is `log_lead_time` and the random-effects structure is $(1 \mid \texttt{seller\_id}) + (1 \mid \texttt{customer\_state})$.

Table 7: Comprehensive model map for the M0–M4 experiment.

Model	Added fixed-effects block	R model configuration (concise)	Sample + estimation	Primary inferential use
M0	Intercept only.	Crossed random-intercept mixed model.	Main sample; REML.	Baseline variance decomposition and ICC reporting (H1).
M1	Routing block: same-state and same-region indicators.	M0 + routing fixed effects.	Main sample; REML + ML refit.	Tests routing contribution via $M0 \rightarrow M1$ LR comparison (H2).
M2	Shipment block: standardized log weight, volume, value, freight, and item count.	M1 + shipment fixed effects.	Main sample; REML + ML refit.	Tests incremental shipment contribution over routing model (H3).
M3	Controls block: season, purchase year, and product-theme factors.	M2 + time and composition controls.	Main sample; REML.	Fully adjusted pooled benchmark for M0–M3 interpretation.
M4_c1	Regime main effects (precluster dummies; $K = 5$).	M3 + regime indicators only.	Cluster-ready sample; REML + ML refit.	Tests whether regime labels add fit beyond pooled M3 ($H_{0,4a}$).
M4_c2	Regime interactions: same-state, freight, and items by precluster.	M3 + regime dummies + selected interactions.	Cluster-ready sample; REML + ML refit.	Preferred heterogeneity model; tests interaction block ($H_{0,4b}$).

Notes: `orders_mod` has n = 93,853 (2,892 sellers; 27 customer states). `orders_mod_cluster` is the common sample used for M3-vs-M4 comparisons with observed `precluster` labels ($K = 5$, treatment coding, cluster 1 reference). REML fits are used for coefficient/variance reporting; ML refits (`REML=FALSE`) are used for nested LR tests and AIC/BIC comparisons. Here $H_{0,4a}$ is "all `precluster` main effects = 0," and $H_{0,4b}$ is "all selected regime-interaction coefficients = 0."

4.5 Robustness and outlier handling

Because lead time retains a heavy right tail even after log transformation, and pre-model MCD diagnostics flag a non-trivial multivariate outlier share (Section 3.3), robustness checks are run as sensitivity analyses around the preferred mixed-model specification (M4_c2). The checks include tail-capped or trimmed outcome variants (including top-tail treatment around the 95th percentile), specification variants (e.g., no-theme and seller-only random effects), and robust flat estimators (OLS, Huber, LTS) to verify sign and relative magnitude stability of key routing and shipment coefficients.(Filzmoser et al., 2008)

For tail sensitivity, a 95th-percentile winsorized outcome is defined as:

$$L_i^{(\text{trim95})} = \min(L_i,\, q_{0.95})\,,$$

where $L_i = \text{lead\_time\_days}_i$ and $q_{0.95}$ is the empirical 95th percentile of lead_time_days in the analysis sample. [4] The robustness criterion for H5 is directional consistency of substantive findings under these alternatives.

4.6 Bootstrap inference for mixed-effects models

To reduce dependence on asymptotic Wald intervals, parametric bootstrap is used on the preferred mixed model.(Davison and Hinkley, 1997)

Bootstrap configuration in this report:

[4] Winsorization caps extremes while retaining all rows; trimming removes rows beyond a threshold and changes the analysis sample.

- **Target model:** the final heterogeneous mixed-effects model M4_c2.
- **Engine/design:** `lme4::bootMer` with `type="parametric"` and `use.u=FALSE`; each replicate simulates a new response from the fitted model and refits the same formula on the same design matrix.
- **Replications/reproducibility:** $B = 80$ refits, with fixed seed (`set.seed(42)`) and serial execution (`parallel="no"`).
- **Tracked coefficients:** same-state main effect, same-region main effect, freight main effect, and two key regime interactions (same-state $\times$ cluster 5; freight $\times$ cluster 5).
- **Interval construction:** 95% percentile intervals from empirical 2.5% and 97.5% quantiles, compared against Wald intervals from the same fitted model.

Here b indexes bootstrap replicate and B is the total number of refits. This $B = 80$ setting is used as a computationally feasible calibration check for large crossed-model refits.(Field and Welsh, 2007)

4.7 Model-based clustering of shipment regimes

Shipment regimes are estimated with Gaussian model-based clustering (`mclust`) on standardized log shipment features: `z_log_wgt`, `z_log_vol`, `z_log_value`, `z_log_freight`, and `z_log_items`.(Fraley and Raftery, 2002) The cluster-engineering workflow has four steps:

1. Fit candidate Gaussian mixtures over $G = 1, \ldots, 10$ and covariance forms `EII`, `VII`, `EEI`, and `VVI` (`E`/`V` denote equal/variable volume-shape assumptions across clusters).
2. Screen K with model-based criteria (Bayesian Information Criterion, BIC; and integrated completed-likelihood, ICL) and a geometry cross-check (silhouette).
3. Apply guardrails over $K \in \{3, 4, 5, 6\}$: assignment uncertainty (posterior entropy proxy), minimum cluster share, and subsample stability (ARI, adjusted Rand index, under repeated subsampling).
4. Select the smallest K that is both stable and operationally non-fragmented; this yields $K = 5$ with covariance form `VII`.

Here G denotes the candidate number of mixture components in screening, K is the finalized regime count used in modelling, and ARI is the adjusted Rand index used for subsample stability checks.

The final `precluster` labels enter M4 in two ways: as regime dummies in M4_c1 and as selected interactions in M4_c2 (notably with routing and freight-related terms) to test heterogeneous sensitivities. These regimes are descriptive heterogeneity controls, not causal latent classes. Selection and profile results are reported in Sections 5.5.1 and 5.5.2; implementation details are summarized in the appendix software notes.

4.8 Diagnostics, model comparison, and linkage to expectations

Model adequacy is evaluated with residual diagnostics (fitted-vs-residual and QQ patterns on the log scale), random-effect diagnostics, and influence checks for extreme orders and dominant clusters. Model comparison uses AIC/BIC, likelihood-ratio tests for nested comparisons, and mixed-model R^2 summaries (R^2_{marg}, R^2_{cond}).(Nakagawa and Schielzeth, 2013)

Table 8: Key mixed-effects diagnostics and how they are interpreted in this study.

Metric	Interpretation in this study
AIC, BIC	Likelihood-based fit criteria with complexity penalty; lower values indicate better fit-complexity balance across nested models.
logLik	Maximised model log-likelihood; higher values indicate better fit and are used for AIC/BIC and likelihood-ratio comparisons.
σ_u^2, σ_v^2, σ_ε^2	Variance components for seller random intercepts, customer-state random intercepts, and residuals; quantify between-seller, between-state, and within-cluster variability in log_lead_time.
ICC shares	$\text{ICC}_{\text{seller}}, \text{ICC}_{\text{state}}, \text{ICC}_{\text{total}}$ from Equations (2) and (3); they measure the share of total variance due to seller effects, state effects, and total clustering.
R^2_{marg}, R^2_{cond}	Marginal and conditional R^2 for mixed models. R^2_{marg} captures fixed-effects explanatory power only, while R^2_{cond} includes fixed + random effects.

Interpretation across the Results section follows the H1–H5 structure directly: H1 from ICC decomposition in M0, H2–H3 from incremental fixed-effect blocks in M1–M3, H4 from regime-conditioned M4 effects, and H5 from robustness plus bootstrap consistency checks.

5 Results

5.1 Overview and roadmap

This section follows the model progression M0–M4 and evaluates H1–H5 with explicit inferential wording where formal tests exist. The flow moves from variance decomposition, to model comparison and progression, to key fixed effects, then to cluster heterogeneity, robustness, and a final hypothesis summary. Full diagnostics and supplementary outputs are kept in the appendix materials.

5.2 Variance decomposition

The results begin with variance decomposition from the null crossed random-intercept model M0:

$$y_{ijk} = \beta_0 + u_j + v_k + \varepsilon_{ijk}.$$

with y_{ijk} as order-level log lead time, β_0 as the grand intercept, u_j the seller random intercept, v_k the customer-state random intercept, and ε_{ijk} the order-level residual (as defined in Section 4.3). Table 9 reports seller variance 0.059, state variance 0.069, and residual variance 0.232, implying $\text{ICC}_{\text{seller}} = 0.164$, $\text{ICC}_{\text{state}} = 0.192$, and $\text{ICC}_{\text{total}} = 0.356$. Around one-third of total variation therefore sits above order level; under the H1 decision rule, this is practical evidence against the no-clustering null.

Model	Seller_Var	State_Var	Residual_Var	ICC_Seller	ICC_State	ICC_Total
M0	0.059	0.067	0.232	0.165	0.186	0.351

Table 9: Variance components and ICC shares for the null crossed random-intercept model (M0; n = 93,853).

Both higher-level ICCs are material, so a single-level model would fold structural seller/state differences into residual noise.

The spread of seller random intercepts around zero indicates meaningful between-seller baseline differences in lead time, consistent with H1.

Figure 11: Distribution of seller random intercepts from M0.

5.3 Model comparison and progression

Table 10 tracks how variance components and fit metrics evolve as fixed-effect blocks are added from M0 to M3. This progression isolates the incremental contribution of routing (M1), shipment covariates (M2), and season/year/theme controls (M3).

Table 10: Variance components and model fit across M0–M3 (same analysis sample).

Model	Seller var	State var	Residual var	$\text{ICC}_{\text{seller}}$	$\text{ICC}_{\text{state}}$	R^2_{marg}	R^2_{cond}
M0: Intercept only	0.059	0.069	0.232	0.164	0.192	0.000	0.351
M1: + routing	0.054	0.043	0.212	0.176	0.139	0.139	0.407
M2: + shipment	0.052	0.035	0.210	0.175	0.116	0.160	0.402
M3: + season/year/themes	0.037	0.032	0.197	0.140	0.119	0.223	0.422

The largest fixed-effect gain appears at M0 → M1, so routing is the strongest added information block. Shipment terms add a smaller increment ($R^2_{\text{marg}} : 0.139 \to 0.160$), and season/year/theme controls add another step-up to 0.223. Even in M3, $\text{ICC}_{\text{total}} = 0.259$, so higher-level heterogeneity is reduced but not eliminated; crossed random effects remain necessary in later models. This makes M3 the pooled benchmark for the fixed-effect and heterogeneity results that follow.

5.4 Key fixed effects

This subsection turns from fit progression to substantive fixed-effect interpretation. The focus is on the pooled M1–M3 sequence, where routing and shipment-related predictors are estimated after accounting for seller and customer-state clustering.

5.4.1 Routing effects (M1–M2)

Routing effects are estimated by adding same-state and same-region indicators in M1 and then controlling for shipment characteristics in M2. Because the outcome is on the log scale, fixed-effect coefficients are interpreted as multiplicative changes in expected lead time through $100 \cdot (\exp(\beta) - 1)$.

Table 11: Routing coefficients in Models M1 and M2 (conditional log-scale estimates and back-transformed percentage effects; stars for direct coefficient significance).

Predictor	M1 estimate	M1 % change	M2 estimate	M2 % change
Same-region (`is_same_region`)	-0.094***	-8.9%	-0.083***	-8.0%
Additional same-state (`is_same_state`)	-0.402***	-33.1%	-0.369***	-30.9%
Implied same-state total ($\beta_{sr} + \beta_{ss}$)	-0.496	-39.1%	-0.452	-36.4%

Notes: stars apply to directly estimated coefficients only ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$); the implied same-state total is a linear combination.

In M2, $\beta_{sr} = -0.083$ implies about 8.0% shorter lead time for same-region (different-state) shipments versus cross-region. The incremental same-state term is $\beta_{ss} = -0.369$; therefore the total same-state contrast versus cross-region is $\beta_{sr} + \beta_{ss} = -0.452$, about 36.4% shorter expected lead time, in line with H2.

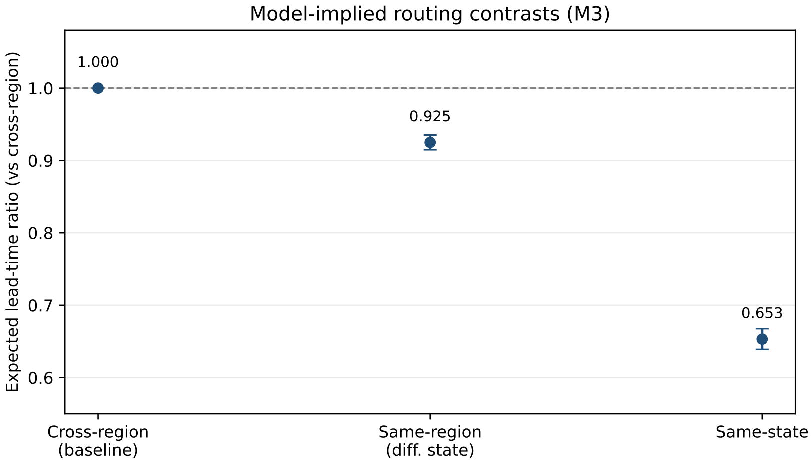


Figure 12: Model-implied routing contrasts from M3 on the original scale (cross-region baseline = 1). Error bars for same-region and same-state are approximate 95% intervals from coefficient CI combinations.

The model-implied routing contrasts show the largest downward shift for same-state shipments, while same-region routing provides a smaller but still clear improvement over the cross-region baseline.
The M0 → M1 likelihood-ratio improvement is large ($\Delta\chi^2 \approx 8358$, $df = 2$, $p < 2.2 \times 10^{-16}$), so $H_{0,2}$ is rejected at conventional levels. After adding shipment covariates in M2, attenuation is modest, leaving routing as the dominant fixed-effect block. Given the nested coding, same-state versus cross-region is interpreted through the combined contrast ($\beta_{sr} + \beta_{ss}$). Figure 12 provides the corresponding model-implied scale interpretation by routing category.

5.4.2 Shipment predictors and time/product controls (M2–M3)

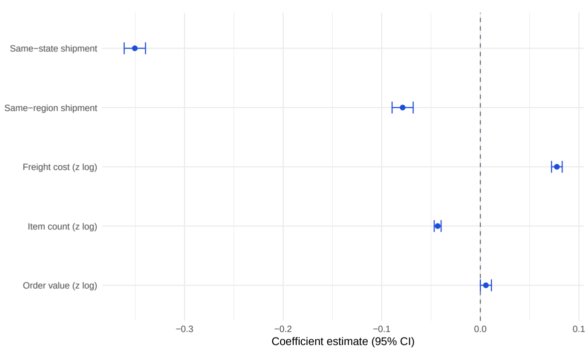
Model M2 adds standardized shipment predictors to routing; Model M3 then adds season, purchase year, and product-theme controls. The summary in Table 12 highlights key terms from the two models using the same log-scale-to-percent interpretation.

Table 12: Shipment and time/product effects in Models M2 and M3

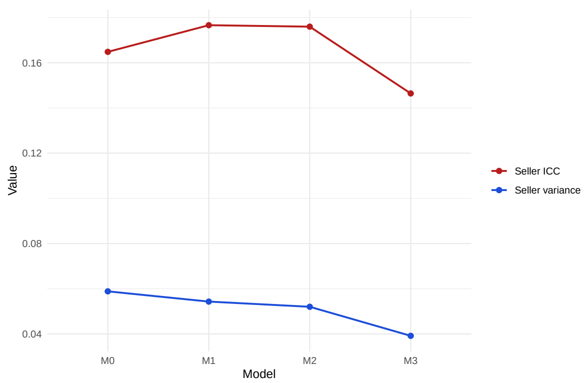
Predictor	M2 estimate	M2 % change	M3 estimate	M3 % change
Shipment predictors (z-log scale)				
Freight cost (`z_log_freight`)	0.056***	+5.8%	0.081***	+8.4%
Item count (`z_log_items`)	-0.036***	-3.6%	-0.044***	-4.3%
Order value (`z_log_value`)	0.009**	+0.9%	0.006*	+0.6%
Weight (`z_log_wgt`)	0.010**	+1.0%	-0.002	-0.2%
Volume (`z_log_vol`)	-0.004	-0.4%	-0.007*	-0.7%
Season effects in M3 (baseline: DJF)				
MAM	–	–	-0.131***	-12.3%
JJA	–	–	-0.348***	-29.4%
SON	–	–	-0.157***	-14.5%
Purchase year 2017 (vs 2016)	–	–	-0.366***	-30.6%
Purchase year 2018 (vs 2016)	–	–	-0.417***	-34.1%
Theme effects in M3 (vs model baseline theme level)				
Experience + Hedonic (baseline)	–	–	(ref.)	–
Experience + Utilitarian	–	–	0.016*	+1.6%
Search + Hedonic	–	–	-0.005	-0.5%
Search + Utilitarian	–	–	0.005	+0.5%

Notes: stars indicate Wald-test significance ($^*p < 0.05$, $^* * p < 0.01$, $^* * *p < 0.001$); non-significant controls are retained for transparency; omitted baseline levels are shown as `(ref.)`.

In M3, freight is the strongest shipment predictor (+8.4% per 1-SD increase), while item count stays negative (-4.3%). Term-level Wald results reject $H_{0,3j}$ for freight, items, and value, while weight/volume are weak after joint control. Season and product-theme factors add control structure (with DJF as the slowest seasonal baseline in this run), yet routing still dominates. From Table 10, M2 $\rightarrow$ M3 increases R^2_{marg} from 0.160 to 0.223 and R^2_{cond} from 0.402 to 0.422, which is consistent with H3.
Figure 13a provides the broader fixed-effect picture for the pooled M3 benchmark.



(a) Selected fixed effects in pooled M3.



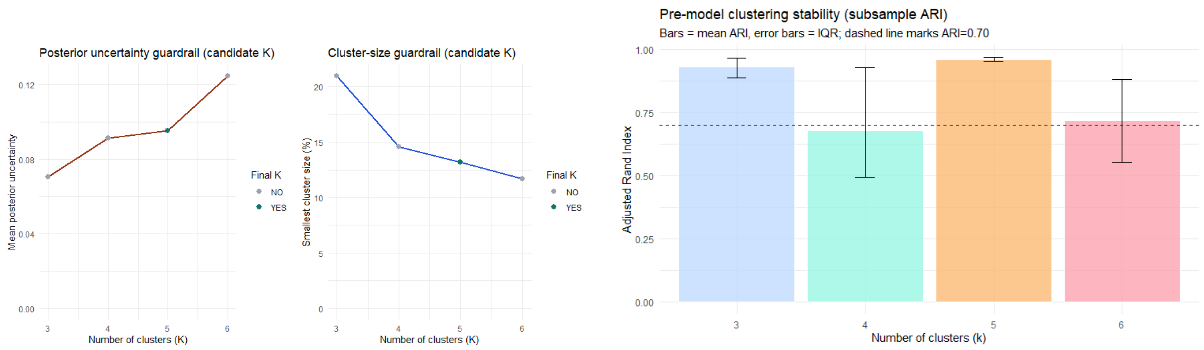
(b) Variance decomposition progression across M0–M3.

Figure 13: Pooled fixed-effect and variance-decomposition views for the M0–M3 sequence.

The visual trajectory matches the tabled results: routing drives the largest early gain, while seller/state clustering persists after pooled controls.

5.5 Cluster-informed heterogeneity (M4: regimes and interactions)

This subsection proceeds in five steps: finalize K, profile clusters, derive H4 expectations, test regime-conditioned models, and evaluate added value versus pooled M3. Regimes are used as descriptive heterogeneity controls, not causal latent classes.



(a) Guardrails by K: assignment uncertainty and minimum cluster share.

(b) Subsample stability by K: adjusted Rand index (ARI).

Figure 14: Cluster finalization diagnostics for selecting K.

5.5.1 Finalize K

Cluster finalization uses a multi-criterion rule, not a single score. I first screen K with model-based criteria (BIC/ICL) and silhouette, then require acceptable assignment certainty, non-fragmented cluster sizes, and high subsample stability before locking the final regime count.

Criterion	Value	Note
BIC (mclust)	10	Boundary optimum on tested grid
ICL (mclust)	10	Boundary optimum on tested grid
Silhouette (kmeans)	3	Geometry-only best k
Candidate K range	3 to 6	Guardrail sensitivity range
Boundary warning	YES	BIC/ICL hit upper G bound
Final K used	5	Stability-priority choice
Selected covariance model	VII	Final mclust covariance form

Table 13: Precluster selection summary and screening outcomes.

BIC/ICL peak at the tested upper bound ($G = 10$), so finalization relies on the guardrail-and-stability stage rather than boundary model-selection maxima.

$K = 5$ satisfies both guardrail conditions and delivers the strongest stability profile among candidate solutions (mean ARI = 0.960, IQR = 0.954 to 0.969).

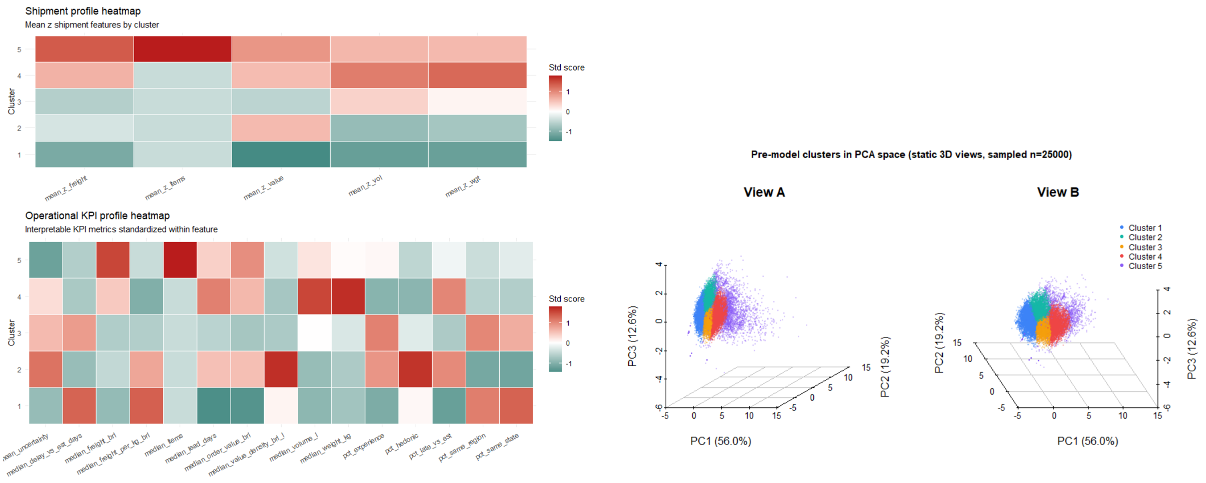
5.5.2 Cluster profiling (shipment and seller-operating patterns)

Precluster	Orders	Share (%)
1	29,846	31.8
2	13,109	14.0
3	23,411	24.9
4	15,120	16.1
5	12,367	13.2

Table 14: Shipment-regime shares in the cluster-ready analysis sample.

Table 14 shows that all five regimes have substantial support (13.2% to 31.8% of orders), so heterogeneity modelling is not driven by very small fragments.

Using Figure 15a (standardized within feature), the cluster patterns are compactly:



(a) Precluster shipment and KPI profile heatmaps.

(b) Three-dimensional PCA projection of finalized regimes ($K = 5$).

Figure 15: Cluster profiling views: standardized KPI signatures and PCA-space separation.

- **C1 (31.8%):** low-load/low-value orders, high same-state and same-region shares, and generally low uncertainty and late-rate.
- **C2 (14.0%):** physically small but value-dense, hedonic-leaning profile with lower local-routing shares and a higher late-rate.
- **C3 (24.9%):** mid-load and comparatively balanced profile, with moderate local-routing shares and intermediate delay behavior.
- **C4 (16.1%):** bulky/heavy profile (higher volume and weight), lower same-state share, and higher late-rate.
- **C5 (13.2%):** high freight, high item count, and higher order value, with lower local-routing shares than C1/C3.

This profile structure shows that clusters differ not only by parcel size but also by routing mix and delivery-risk indicators.

The PCA geometry shows partial but clear separation: regimes overlap, yet occupy different high-density zones, so they are treated as operational regimes rather than hard classes.

Together, Figures 15a and 15b supports H4: coefficient directions may stay stable, but effect magnitudes can differ by regime.

5.5.3 Cluster interaction in modelling

Because the regimes differ in operational shipment context, the pooled-slope assumption in M3 is likely too restrictive. The profile evidence motivates H4: routing and freight sensitivities should vary in magnitude across regimes, while preserving economically coherent signs.

I test this by extending M3 in two stages on the common cluster-ready sample: M4_c1 adds regime main effects, and M4_c2 adds selected regime-conditioned interactions for routing and shipment terms.

Model	AIC	BIC	logLik	R^2_{marg}	R^2_{cond}
M3 pooled baseline	117189	117369	-58576	0.222	0.424
M4_c1 (add precluster main effects)	117138	117356	-58546	0.223	0.424
M4_c2 (add precluster interactions)	115806	116099	-57872	0.247	0.428

Table 15: Model comparison for the cluster-conditioned extension on the common cluster-ready sample (same random-effects structure across models).

Table 15 indicates that M4_c1 improves fit over M3 ($\chi^2 = 58.95$, $df = 4$, $p = 4.83 \times 10^{-12}$), so $H_{0,4a}$ is rejected. Adding regime-specific interactions in M4_c2 yields a much larger gain ($\chi^2 = 1348.1$, $df = 8$, $p < 2.2 \times 10^{-16}$), so $H_{0,4b}$ is also rejected. AIC/BIC drop materially (117189/117369 $\rightarrow$ 115806/116099), while R^2_{marg} rises from 0.222 to 0.247 and R^2_{cond} from 0.424 to 0.428. In short, heterogeneity terms add real signal rather than relabeling, so M4_c2 is retained as the preferred heterogeneous specification and M3 as the pooled benchmark.

Regime	Same-state coef.	Same-state %	Freight coef.	Freight %
C1	-0.187 [-0.206, -0.169]	-17.1%	0.326 [0.310, 0.341]	+38.5%
C2	-0.297 [-0.346, -0.250]	-25.7%	0.158 [0.118, 0.196]	+17.1%
C3	-0.291 [-0.335, -0.249]	-25.2%	0.183 [0.143, 0.222]	+20.1%
C4	-0.335 [-0.380, -0.292]	-28.5%	0.053 [0.018, 0.087]	+5.4%
C5	-0.383 [-0.427, -0.339]	-31.8%	0.045 [0.012, 0.077]	+4.6%

Table 16: Implied regime-specific same-state and freight effects under M4_c2.

Notes: entries are derived contrasts (baseline + interaction terms), so star markers are not shown; practical significance is assessed by whether the reported 95% interval excludes zero.

Same-state effects are negative and freight effects are positive in all five regimes; heterogeneity appears in effect size rather than sign, matching the core H4 pattern.

For Table 16, regime coefficients are formed as baseline + interaction terms from M4_c2; the displayed interval bounds are approximate combinations from reported Wald limits.

Table 16 points to material heterogeneity with stable directions. In cluster-stratified mixed refits, same-state coefficients are negative in every regime (approximately -0.355 to -0.189), while freight coefficients are positive (approximately 0.034 to 0.399). This supports H4 and shows that pooled M3 masks regime-level magnitude differences.

5.6 Robustness and uncertainty

This subsection tests whether the main conclusions from M3 and M4_c2 are sensitive to extreme delays, specification choices, and interval estimation. The robustness battery combines trimming/specification variants, robust estimators, regional submodels, and parametric bootstrap intervals.

Trimming and specification variants. I first evaluate tail and structure sensitivity via trim95 winsorization, no-theme and seller-only mixed-model variants, and regional refits. Core routing and shipment terms remain directionally stable across trim95 and structure variants, indicating that the main inferences are not driven by a small set of extreme observations.

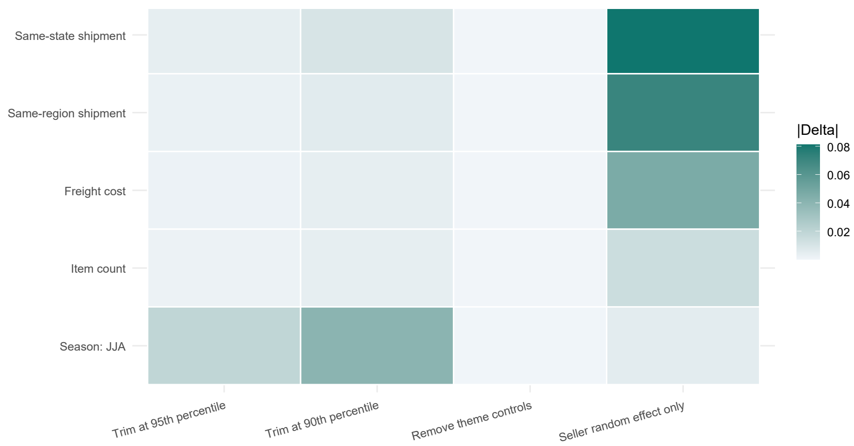


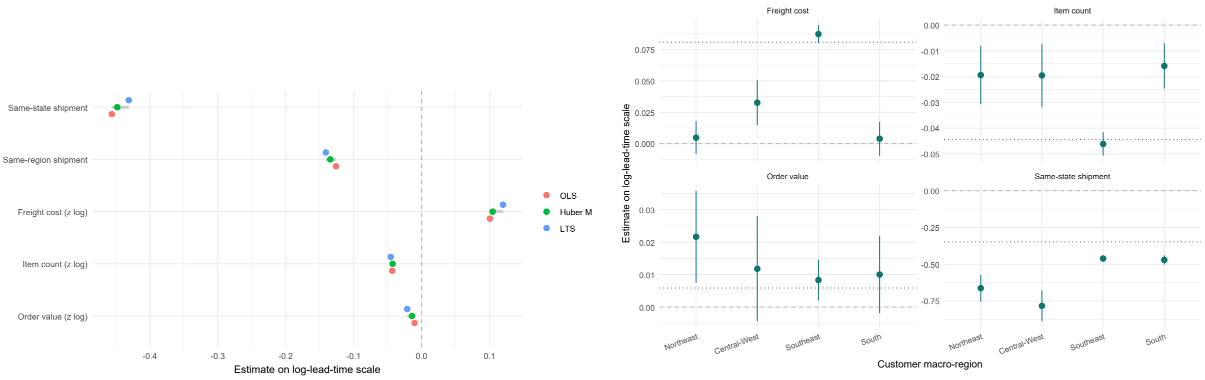
Figure 16: Sensitivity heatmap: absolute coefficient change versus M4_c2 across trim95 and structure variants.

Most cells remain low-intensity, indicating modest drift of key coefficients across robustness variants.

Term	M4_c2	Trim95	$\|\Delta\|$	No-theme	$\|\Delta\|$	Seller-only $\|\Delta\|$
Same-state shipment	-0.187	-0.183	0.004	-0.187	0.000	0.024
Same-region shipment	-0.066	-0.064	0.002	-0.066	0.000	0.058
Freight cost z(log)	0.326	0.328	0.002	0.326	0.000	0.091
Item count z(log)	-0.034	-0.031	0.003	-0.034	0.000	0.012
Same-state $\times$ cluster 5	-0.196	-0.198	0.002	-0.196	0.000	0.079
Freight $\times$ cluster 5	-0.281	-0.285	0.004	-0.281	0.000	0.070
Season: JJA	-0.352	-0.333	0.019	-0.352	0.000	0.007

Table 17: Coefficient stability across M4_c2 variants with absolute deltas versus reference model.

Table 17 quantifies the same pattern numerically: coefficient drift is small in trim95 and no-theme variants, with larger but still direction-preserving shifts in the seller-only structure.



(a) Classical vs robust coefficient estimates.

(b) Regional mixed-model estimates vs national reference.

Figure 17: Robustness triangulation across estimator type and macro-regional refits.

Across these checks, no key sign reversals appear for routing, freight, or highlighted interaction terms. Magnitude shifts are moderate (largest mixed-model drift about 0.091), satisfying the H5 robustness criterion.

Robust estimators and regional transportability. Figure 17a shows that robust estimators (Huber/LTS) produce very similar directions and close magnitudes to classical estimates for the key terms. Figure 17b further indicates that the national sign pattern is preserved in regional mixed refits, with variation mainly in effect size rather than direction.

Bootstrap versus Wald intervals. I then validate interval uncertainty with parametric bootstrap for M4_c2. The run settings are:

- parametric bootstrap via `bootMer` (`use.u=FALSE`);
- $B = 80$ bootstrap refits with `set.seed(42)`;
- 95% percentile intervals for the selected routing/freight terms and key interactions;
- direct comparison against Wald intervals for the same terms.

term	point_estimate	boot_perc_low	boot_perc_high	wald_low	wald_high	interval_overlap
Same-state shipment	-0.348	-0.361	-0.337	-0.359	-0.337	TRUE
Same-region shipment	-0.078	-0.091	-0.068	-0.089	-0.067	TRUE
Freight cost	0.081	0.076	0.086	0.075	0.086	TRUE
Item count	-0.044	-0.047	-0.041	-0.048	-0.041	TRUE
Season: JJA	-0.348	-0.355	-0.339	-0.356	-0.339	TRUE

Table 18: Bootstrap percentile intervals ($B = 80$) vs Wald intervals for selected M4_c2 parameters.

Table 18 shows close alignment between bootstrap and Wald intervals for key pooled and interaction terms, so inferential conclusions are unchanged by resampling-based uncertainty checks.

Across trimming, robust estimators, regional submodels, and bootstrap inference, the directional ordering of effects is unchanged: routing is dominant, shipment covariates provide smaller but stable refinements, and seller/state clustering stays material.

5.7 Summary relative to hypotheses

The combined evidence answers RQ1–RQ3 coherently: variance is decomposed across sellers and states, routing is the dominant fixed-effect driver with shipment-level refinements, and the main conclusions remain stable under multiple stress tests.

Table 19: Summary of empirical support for H1–H5.

Hyp.	Status	H_0 decision	Evidence (section)
H1	Yes	Practical rejection of no-clustering null	M0 shows material clustering ($\text{ICC}_{\text{seller}} \approx 0.164$, $\text{ICC}_{\text{state}} \approx 0.192$); M3 still has $\text{ICC}_{\text{total}} = 0.259$ (Sections 5.2 and 5.3).
H2	Yes	Reject $H_{0,2}$	Routing block LR test $M0 \to M1$: $\Delta\chi^2 \approx 8358$, $df = 2$, $p < 2.2 \times 10^{-16}$; same-state and same-region effects are strongly negative (Section 5.4.1).
H3	Largely yes	Partial rejection (term-level)	Wald tests reject term-level nulls for freight/items/value, while weight/volume are weaker under joint control; shipment block remains smaller than routing (Section 5.4.2).
H4	Yes	Reject $H_{0,4a}$, reject $H_{0,4b}$	LR tests: $M3 \to M4\_c1$ ($p = 4.83 \times 10^{-12}$) and $M4\_c1 \to M4\_c2$ ($p < 2.2 \times 10^{-16}$); regime magnitudes differ with stable signs (Section 5.5).
H5	Yes	Not a single NHST null	Trimming, robust estimators, regional refits, and bootstrap checks preserve the main directional conclusions (Section 5.6).

The evidence supports all five directional expectations, with H3 classified as “largely yes” because shipment effects are meaningful but clearly smaller than routing effects.

6 Discussion

6.1 Summary of key findings

This section revisits the three research questions from the Introduction and summarizes how the empirical results in Section 5 address them.

For RQ1 (variance decomposition), the null model M0 and the fully adjusted model M3 both show substantial higher-level heterogeneity. In M0, around 16% of variance in log lead time is between sellers and 19% between customer states, with about 64% at order level (Section 5.2). After adding routing, shipment, and time/product controls, M3 still leaves $\text{ICC}_{\text{total}} = 0.259$ (Section 5.3). This is consistent with H1 and confirms meaningful clustering by sellers and destinations.

For RQ2 (key drivers), routing is the dominant block in all pooled models. Same-state shipping is associated with roughly 30–40% shorter expected lead time than cross-region shipping, while same-region (different-state) shipments are about 7–10% faster (Section 5.4.1). Shipment covariates add smaller but systematic refinements: higher freight is associated with longer lead time, higher item count with slightly shorter lead time, and order value with a small positive association (Section 5.4.2). These patterns support H2 and largely support H3.

For RQ3 (robustness and stability), heterogeneity and robustness analyses point to magnitude variation with stable directions. In cluster-stratified mixed refits, same-state effects remain negative across regimes (about -0.355 to -0.189) and freight effects remain positive (about 0.034 to 0.399) (Section 5.5), supporting H4. Trimming, robust estimators, regional refits, and bootstrap-vs-Wald intervals preserve the directional ordering of the main effects with modest drift (Section 5.6), supporting H5.

6.2 Interpretation and theoretical implications

The results suggest that delivery performance is a structural coordination problem, not only a parcel-level prediction problem. Persistent seller and geography effects indicate stable differences in fulfillment capability and regional frictions, consistent with operations and service-quality perspectives in e-commerce logistics (Fleischmann et al., 2008; Gunasekaran et al., 2002; Boyer et al., 2005). The strong same-state and same-region effects indicate that spatial matching between order demand and fulfillment location is a major determinant of service performance.

The findings also align with evidence that delivery reliability is central to customer outcomes (Xiao et al., 2017). Relative to prior studies focused on pooled average effects, this report extends the literature in three ways: a multilevel decomposition across orders, sellers, and destinations (RQ1/H1), a clear routing-dominance result after shipment/time/product controls (RQ2/H2–H3), and regime-dependent effect magnitudes with stable signs (RQ3/H4–H5).

6.3 Methodological and AMS implications

Methodologically, this study shows why a combined AMS workflow is useful for business data with hierarchy, heavy tails, and latent heterogeneity. Crossed mixed-effects models provide the main inferential backbone for dependence and variance decomposition (Gelman and Hill, 2007). Robust estimators and trimming address tail sensitivity (Filzmoser et al., 2008); in shipment-feature space, pre-model MCD diagnostics (Section 3.3) flag roughly 9.6% of orders as multivariate outliers under a 97.5% cutoff, consistent with heavy-tailed structure. Bootstrap intervals validate uncertainty under unbalanced clusters (Davison and Hinkley, 1997), and model-based clustering introduces interpretable regime heterogeneity (Fraley and Raftery, 2002).

The practical AMS lesson is to treat these tools as complementary safeguards rather than substitutes. In this application, multilevel models answer RQ1–RQ2, while robustness and bootstrap modules stress-test RQ3. Each method targets a different failure mode (dependence, outliers, asymptotic fragility, latent segmentation), and their joint use improves inferential credibility.

6.4 Limitations

- **Data scope:** the sample covers one marketplace and one historical window (Brazil, 2016–2018), so generalization to other platforms, periods, or countries is limited.
- **Measurement scope:** explicit route distance, carrier identity, warehouse process metrics, and contract terms are not observed; part of their influence is likely absorbed into route indicators and seller/state random effects.
- **Model structure:** the main models are primarily random-intercept and static; additional random slopes or dynamic time structure could capture further heterogeneity.
- **Identification:** the design is observational, so reported effects are associative rather than causal and may still reflect omitted operational confounding.

6.5 Directions for future research

Three extensions are especially relevant. First, enrich the data with carrier- and route-level variables (distance, service tier, hub topology) to separate routing and process channels more directly. Second, estimate dynamic multilevel models with explicit time effects and random slopes (e.g., for same-state or freight) to capture evolving logistics conditions. Third, test transportability and identification through cross-market replication and quasi-experimental designs around policy or network changes, while extending outcomes to delay-vs-promise, satisfaction, and cancellations.

7 Conclusion

This report examined delivery lead-time variation in Brazilian e-commerce using an integrated AMS pipeline. Empirically, it finds meaningful seller/state clustering (RQ1/H1), dominant routing effects (RQ2/H2), secondary but stable shipment refinements (RQ2/H3), and regime-dependent heterogeneity that remains directionally robust under stress tests (RQ3/H4–H5). Methodologically, it shows that combining multilevel modelling, robust sensitivity analysis, bootstrap inference, and model-based clustering provides a transparent framework for hierarchical, heavy-tailed business data. The main operational implication is to manage route design and geography as first-order levers, while using shipment regimes and shipment covariates for more granular estimated delivery-time (ETA) and service-policy design.

8 Appendix

Software and implementation notes

- Core software stack: R with `lme4`, `lmerTest`, `performance`, `robustbase`, `MASS`, and `mclust`, plus tidyverse utilities for preprocessing and reporting.
- The full execution workflow is scripted in `code/main-code-v3.Rmd`, including data locking, M0–M4 estimation, robustness variants, regional refits, and bootstrap inference.

1. Build the strict modeling dataset (n = 93,853) with transformed routing, shipment, and control variables.
2. Fit nested crossed mixed-effects models M0–M3 and extract variance and fixed-effect summaries.
3. Fit M4_c1 and M4_c2 on the cluster-ready sample, then evaluate regime heterogeneity.
4. Run robustness variants (trim95, structure variants, robust estimators, regional submodels).
5. Run parametric bootstrap for selected M4_c2 parameters and compare percentile versus Wald intervals.

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